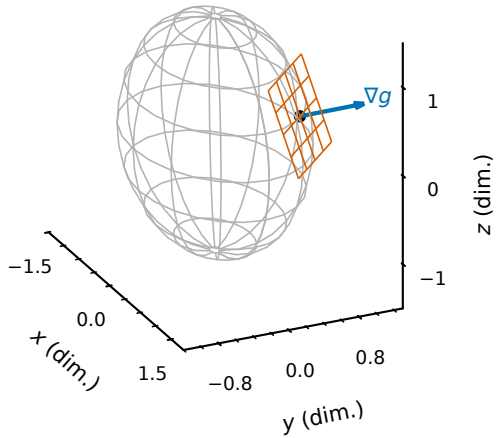
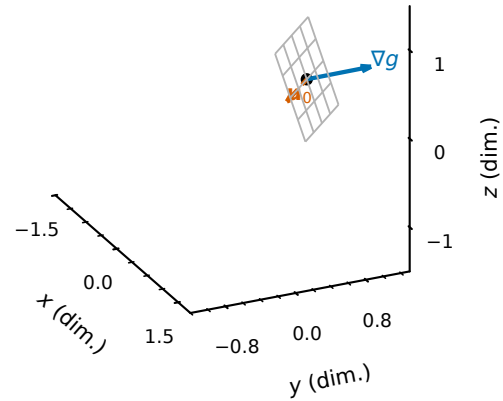
A



Input space: $g = x^2 + 2y^2 + z^2 = 7/4$

B



$\nabla g = (2, 2, 1)^T, \quad D_{\mathbf{u}_0} g = 0$